

Jiessie Tie

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EDUCATION

UNIVERSITY OF TORONTO ST. GEORGE, TORONTO, ON, CANADA

Doctor of Philosophy (PhD) Expected Graduation 2029 — cGPA: 4.0/4.0

- Advisor: Dr. Shurui Zhou (*ForColab*)

Masters of Science (MScs.) Graduation 2026 — cGPA: 3.9/4.0

- MSc Paper: "*Should I Give Up Now?*" Investigating LLM Pitfalls in Software Engineering

- Advisor: Dr. Shurui Zhou (*ForColab*)

Honours Bachelor of Science (HB.Sc.) Graduation 2024 — cGPA: 3.81/4.0

- Cognitive Science, Computer Science, Psychology

RESEARCH EXPERIENCE

UOFT FORCOLAB

2023/PRESENT

PhD Student, supervised by Dr. Shurui Zhou

- **Project: LLM-Supported Clinical Pathway Generation (Industry Collaboration: Verto Health).**

Designed LLM-based systems for generating patient-specific clinical pathways within real-world healthcare workflows.

- **Project: LLM-Mediated Collaboration in Hackathons.**

Conducted a mixed-methods study of how teams appropriate LLMs for coordination and decision-making.

Publication: Tie J, Pudari R, Cheng J, Guo J, Zhou, S. (2025). "Can You Ask the LLM?": Understanding LLM-Mediated Collaboration in a Hackathon. (*In Submission*)

- **Project: LLM Failures in Software Engineering.**

Empirically analyzed LLM failure modes and developer recovery strategies in software engineering tasks.

Publication: Tie J, Yao B, Li T, Ahmed S. I, Wang D, Zhou S (2024). "LLMs are Imperfect, Then What? An Empirical Study on LLM Failures in Software Engineering". (TOSEM)

- **Project: LLMs in Scientific Programming Tasks**

Studied how scientists use LLMs and AI-assisted coding tools for domain-specific programming.

Poster: LLMs in Scientific Programming Tasks. DSI Talent Showcase (2024)

UOFT COMPUTATION SOCIAL SCIENCE LAB

2023/2024

Research assistant, supervised by Dr. Ashton Anderson

- **Project: Algorithm Aversion in Large Language Models.**

Conducted a behavioural evaluation of LLMs to study algorithm aversion and appreciation, building experimental pipelines with LangChain and the OpenAI API and analyzing results with statistical methods; findings inform AI interpretability, trust, and human-AI interaction.

Publication: Bo J*, Mok L*, Tie J, Anderson A (2024). "Does GPT Distrust Algorithms? Evaluating Large Language Models for Algorithm Aversion". *CHI '24 HEAL Workshop* and *IC2S2 '24*.

UOFT MIND AND BODY LAB

2023/2024

- Applied natural language processing techniques, including topic modelling and semantic analysis, to analyse morality scores in political news articles using the Moral Foundations Dictionary.
- Research informs our understanding of morality shifts and their implications for media and political discourse.
- Created a novel methodology for scoring word morality scores based on perceived political sentiment analysis.

TMU MULTI-MEDIA LAB

2022-2024

- Leveraged proficiency in database management, including MySQL and MongoDB, to support data modelling, query processing, transaction management, and design.
- Successfully utilised various machine learning models (RNN, LSTM, GAN) to enhance research designs and analysed and visualised data using Python libraries (Pandas, NumPy, Seaborn, Plotly, Scikit-learn).

PEKING UNIVERSITY EXCHANGE UNDERGRAD RESEARCHER

2020/2022

- Improved multitrack music generation using MTT-GAN, working with MIDI files, using Python, and Java.
- Developed a Spring Boot application that uses OpenCV and SQL databases to generate musical sequences based on facial recognition.
- Implemented APIs for server-side interface design and OpenCV for enhancing music generation variability, and applied results to enhance musical therapy software with GPT-3

PUBLICATIONS

- **Tie J**, Pudari R, Cheng J, Guo J, Zhou, S. (2025). "Can You Ask the LLM?": Understanding LLM-Mediated Collaboration in a Hackathon. (*In Submission*)
- **Tie J**, Yao B, Li T, Ahmed S. I, Wang D, Zhou S (2024). "LLMs are Imperfect, Then What? An Empirical Study on LLM Failures in Software Engineering". *ACM Transactions on Software Engineering and Methodology* (TOSEM) 2026.
- Bo J*, Mok L*, **Tie J**, Anderson A (2024). "Does GPT Distrust Algorithms? Evaluating Large Language Models for Algorithm Aversion". *CHI '24 HEAL Workshop and IC2S2 '24*.
- Yun T, Guo X, Zhang D, **Tie J**, Qi L, Lu Y (2024). "Hybrid Learning Module-Based Transformer for Multitrack Music Generation With Music Theory". *IEEE Transactions on Computational Social Systems*, doi: 10.1109/TCSS.2024.3486604.
- Yun T, Shuo W, Li X, Huijie C, **Tie J**, Lin Q. (2023). "Spatio-temporal Network for Emotion Recognition in User-Generated Videos". *IEEE Transactions on Network Science and Engineering*
- Yun T, Li X, Zhang T, Jin C, Zhao X, **Tie J**. (2022). "Deep learning based audio and video cross-modal recommendation". *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. doi: 10.1109/SMC53654.2022.9945521.
- Jin C, Wang T, Li X, **Tie J**, Yun T, Liu S, Yan M, Li Y, Wang J, Huang S (2021). "A transformer generative Adversarial Network for multi-track music generation". *CAAI Transactions on Intelligence Technology*, 1-12. <https://doi.org/10.1049/cit2.12065>

PRESENTATIONS

- **Data Sciences Institute Talent Showcase**. "LLMs in Scientific Programming Tasks." (2024)

TEACHING EXPERIENCES | UofT

Coordinated teaching materials, assessments, held tutorials, and office hours for:

· ECE444: Software Engineering (Lead TA)	2 Semesters
· CSC343: Introduction To Databases	3 Semesters
· CSC207: Software Design	3 Semesters
· CSC148: Introduction To Python (Lead TA)	6 Semesters
· CSC108: Introduction To Computer Programming	2 Semesters

INDUSTRY EXPERIENCES

RESEARCH ASSISTANT | VERTO HEALTH 2025/2026

- Designed LLM systems for patient-specific clinical pathway generation in healthcare workflows.

COMPUTER SCIENCE INSTRUCTOR | STEAMBOAT FE 2023/2024

- Taught Python and programming structure to students, involving curriculum for Leetcode, Kaggle and CEMC

VOLUNTEERING & OUTREACH

MENTOR| GRADUATE APPLICATION ASSISTANCE PROGRAM (GAAP) 2025

- Mentor for prospective applicants

EXECUTIVE INSTRUCTOR| MUSICALMINDS 2022/2024

- Organised and led the piano instructor team to help improve musical education in communities of need.

COMPUTER SCIENCE INSTRUCTOR | PROJECT INCLUDE 2020/2021

- Led development on front-end website creation using **React JS**, **HTML**, and **CSS**, and created a personalized **JavaScript** library, teaching a personalized curriculum in workshops and boot camps.

ACADEMIC SERVICE

STUDENT VOLUNTEER| ICSE 2025

- Supported ICSE conference activities by assisting with session logistics and on-site operations.

WORKSHOP ORGANISER | RESPONSIBLE AI HACKATHON, UOFT 2025

- Organized and facilitated a university-wide Responsible AI hackathon in collaboration with Schwartz Reisman Institute for Technology and Society (SRI) and the Data Sciences Institute (DSI), coordinating interdisciplinary teams and research-focused activities.

AWARDS

- Internal Award, Department of Computer Science, UofT, 2025 (\$800)
- Entrance Scholarship, Department of Computer Science, UofT, 2024 (\$10,000)
- Arts & Science Scholarship for Excellence in Cognitive Science, UofT, 2022 (\$1,470)
- Dean's List Scholar, 2020-2023